

```
1  #include <stdio.h>
2  #include <stdbool.h>
3
4  #define MAX 20
5
6  int board[MAX][MAX];
7  int n;
8
9  bool isSafe(int row, int col) {
10     int i, j;
11
12     for (i = 0; i < row; i++)
13         if (board[i][col])
14             return false;
15
16     for (i = row - 1, j = col - 1; i >= 0 && j >= 0; i--, j--)
17         if (board[i][j])
18             return false;
19
20     for (i = row - 1, j = col + 1; i >= 0 && j < n; i--, j++)
21         if (board[i][j])
22             return false;
23
24     return true;
25 }
26
27 bool solve(int row) {
28     if (row == n)
29         return true;
30
31     for (int col = 0; col < n; col++) {
32         if (isSafe(row, col)) {
33             board[row][col] = 1;
34
35             if (solve(row + 1))
36                 return true;
37
38             board[row][col] = 0;
39         }
40     }
41     return false;
42 }
```

```
43
44 void printBoard() {
45     for (int i = 0; i < n; i++) {
46         for (int j = 0; j < n; j++) {
47             printf(board[i][j] ? "Q " : ". ");
48         }
49         printf("\n");
50     }
51 }
52
53 int main() {
54     printf("Enter N: ");
55     scanf("%d", &n);
56
57     if (solve(0)) {
58         printf("\nSolution:\n");
59         printBoard();
60     } else {
61         printf("No solution\n");
62     }
63
64     return 0;
65 }
66
```

Enter N: 8

Solution:

Q	.	.	.	.	.	.	.
.	.	.	.	Q	.	.	.
.	.	.	.	.	.	.	Q
.	.	.	.	.	Q	.	.
.	.	Q	.	.	.	.	.
.	.	.	.	.	.	Q	.
.	Q	.	.	.	.	.	.
.	.	.	Q	.	.	.	.

Process returned 0 (0x0)

Press any key to continue.

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